Part A: Cryptographic Primitives & Key Exchange

Question 1: Explain the mathematical basis of Diffie-Hellman Key Exchange.

Consider prime $p = 23$ and primitive root $g = 5$.

Alice selects private key $a = 6$. Calculate her public key A.

Bob selects private key $b = 15$. Calculate his public key B.

Compute the shared secret key s computed by both parties.

Explain how Man-in-the-Middle (MitM) attacks threaten unauthenticated DH.

Part B: Block Ciphers & Modes of Operation

Question 2: Compare ECB mode versus CBC and GCM modes in AES-256.

Why does Electronic Codebook (ECB) fail semantic security for structured images?

Describe the initialization vector (IV) requirements in Cipher Block Chaining.

Explain how Galois/Counter Mode (GCM) provides both confidentiality and authentication.

Demonstrate how bit-flipping attacks in CBC mode can alter plaintext.

Part C: Web Security & Access Control

Question 3: Distinguish between Cross-Site Scripting (XSS) and CSRF.

Provide a realistic scenario where an attacker leverages Reflected XSS to steal cookies.

What defensive properties does the SameSite=Strict cookie attribute confer?

How does Content Security Policy (CSP) mitigate malicious script execution?

Describe Role-Based Access Control (RBAC) vs Attribute-Based Access Control (ABAC).